Government of Maharashtra
GOVERNMENT POLYTECHNIC, NAGPUR
(An Autonomous Institute of Govt. of Maharashtra)
Near Mangalwari Bazar, Sadar, Nagpur-440001

COURSE CURRICULUM

Program	:	Diploma in CM
Course Category	:	DSC
Course Code	:	CM205H
Course Title	:	JAVA PROGRAMMING

I Rationale:

The Java Programming course provides students with essential, in-demand skills for the software industry. Widely used in enterprise applications, Android development, and web services, Java offers a strong, object-oriented foundation for understanding complex programming concepts. This course enhances problem-solving abilities and prepares students for diverse roles, aligning with industry needs and boosting employability.

II Industry Identified Competency:

The Student will be able to do in Industry at entry level:

Demonstrate proficiency in Java Programming

III Course Outcomes (COs):

After completing this course students will be able to:

CO1: **Apply** the fundamental principles of Java Programming to construct an application.

CO2: **Analyze** and implement basic features of Object Oriented Programming to architect an application design.

CO3: **Develop** java programs incorporating classes, inheritance, interfaces and packages to demonstrate proficiency.

CO4: **Employ** advanced error handling techniques.

CO5: **Implement** multithreaded programming practices to engineer Parallel Processing Applications.

CO6: **Demonstrate** proficiency in Java's I/O operations by employing streams effectively.

IV Learning Scheme:

Classroom Learning (CL)	Tutorial Learning (TL)	Laboratory Learning (LL)	Self-Study Learning (SL)	Notional Learning Hours (NLH)	Credits
4	-	2	2	8	4

V Assessment Scheme:

Classroom Learning				Tutorial/Laboratory Learning				Self-Study Learning	
FA	SA	Total		FA		SA		SA	
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min
30	70	100	40	25	10	25+	10	25	10

VI Classroom Learning Content:

Unit No	Specific Learning Outcomes (SLO) (In Cognitive Domain)		Topics and Sub-topics		Aligned Cos
gramming	1a	Identify key features of Java.	1.1	Java Fundamentals: Evolution of Java, its key features, and advantages, Java Virtual Machine (JVM), Java bytecode	
	1b	Utilize variable management in java applications.	1.2	Primitive data types in java: byte, short, int, long, float, double, char and boolean, Understanding different types of literals: Integer, Floating-Point, Boolean, Character, String, Null. Variable Management: Learning to declare variables, dynamically initialize them, and comprehend their scope and lifetime	

UNIT I: Foundations of Java Proc	1c	Apply type conversion across diverse programming scenarios.	1.3	Type Conversion and Operators: Java's automatic conversions, casting incompatible types, the type promotion rules. Types of operators: Arithmetic, Bitwise, Relational, Logical, Assignment, The ? : operator, operator precedence & associativity, Mathematical functions: min( ), max( ),sqrt( ), pow( ), exp( ), round( ), abs( )	CO1
	1d	Apply control flow in java to solve given problem.	1.4	Control flow in Java: Selection Statements - if, switch, Iteration Statements- while, do-while, for, for-each version of for loop, nested loops, labeled loops, jump statements - break, continue	
	1e	Utilize Java arrays effectively to develop versatile software.	1.5	Arrays in Java: One dimensional, multidimensional, using command line arguments	
: Java Object Oriented Programming Essentials	2a	Create object using new keyword.	2.1	Foundations of Object Oriented Programming: Class fundamentals, object creation with new keyword, reference variables	CO2
	2b	Implement access control mechanisms using Java's access specifiers.	2.2	Visibility and Access Control: Access specifiers - public, private, protected, default	
	2c	Apply given method functionalities in Java.	2.3	Methods and Functionality: Introducing methods, parameterized methods and varargs (variable length arguments), methods returning a value, methods with object as parameter, methods returning an object, method overloading	
	2d	Utilize constructors in Java to accommodate diverse object creation	2.4	Introducing constructors, types of constructors, constructor overloading	

UNIT II	2e	Apply given Java concepts and features.	2.5	Key concepts and utilization of keywords : static, this, final. Garbage collection in Java, utilization of String class, Wrapper classes, concept of nested and inner class	
UNIT III: Inheritance, Polymorphism and Package Management	3a	Apply the appropriate form of inheritance to address a specific programming scenario.	3.1	Inheritance: Inheritance basics, types of inheritance, member access and inheritance, referencing subclass object with superclass reference, using super, creating multilevel hierarch, constructor invocation in multilevel inheritance hierarchy	CO3
	3b	Apply method overriding and dynamic method dispatch to achieve polymorphic behavior.	3.2	Method overriding, Dynamic method dispatch, applying method overriding	
	3c	Utilize abstract classes for defining common interfaces.	3.3	Using abstract classes, using final with inheritance(to prevent overriding and to prevent inheritance), The Object class	
	3d	Construct flexible application by defining and implementing interfaces.	3.4	Interfaces: Defining an interface, implementing interfaces, nested interfaces, applying interfaces, variables in interfaces, extending interfaces, interface references	
	3e	Design organized and secure Java applications by defining and managing packages.	3.5	Package: Defining a package, finding packages and CLASSPATH, access protection, importing packages	
ng	4a	Develop robust and error-resistant applications by mastering exception handling fundamentals.	4.1	Exception Handling Fundamentals, Exception types, uncaught exceptions	

UNIT IV: Exception Handli	4b	Implement error handling in software applications by using try and catch blocks.	4.2	Using try and catch, displaying a description of an exception, multiple catch clauses, Nested try statements, throw, throws, finally	CO4
	4c	Utilize Java's built-in exceptions and custom exception subclasses to develop reliable application.	4.3	Java's Built-in Exceptions, creating user defined Exception subclasses	
	4d	Implement chained exceptions and utilize advanced Exception handling features.	4.4	Chained Exceptions, try-with-resources, multi-catch, final rethrow	
UNIT V: Multithreaded Programming	5a	Implement concurrent Java applications using the Thread class and Runnable interface.	5.1	The Java Thread Model, Thread class and Runnable interface, the Main thread, creating a thread by implementing Runnable, by extending Thread	CO5
	5b	Apply proper thread synchronization and coordination among multiple threads for given scinario.	5.2	Creating multiple threads, using isAlive() and join(),using synchronized methods, the Synchronized statement	
	5c	Implement effective inter-thread communication to avoid deadlock.	5.3	Interthread communication, deadlock, suspending, resuming and stopping thread	
	5d	Manage thread behavior in Java by obtaining thread states.	5.4	Obtaining thread's state, using multithreading	
VI: Java File Handling	6a	Develop robust Java applications by understanding and utilizing I/O basics.	6.1	I/O basics, The predefined streams, Character and Byte streams, Readers and Writers	CO6
	6b	Employ diverse range of I/O operations in Java using Streams.	6.2	Reading console input, reading characters, strings, Writing console output, PrintWriter class, Scanner class, data streams, Serialized Object I/O	

UNIT	6c	Utilize Java's file I/O operations for seamless data management.	6.3	Reading and Writing files, automatically closing a file, copying a file, storing objects in files

VII Laboratory Learning Content:

Unit No	Specific Learning Outcomes (SLO) (In Psychomotor Domain)		Hours	Aligned COs
I	1	Install Java development environment setup	2#	CO1
I	2	Develop programs to demonstrate operators, variables, constant	2	CO1
I	3	Develop programs to demonstrate use of control statements for branching and looping	2#	CO1
I	4	Develop programs to demonstrate Arrays(1D,2D) and command line arguments	2#	CO1
II	5	Develop java programs to demonstrate class, object, methods (including parameterized methods, methods returning value, object), method overloading, constructor overloading	2#	CO2
II	6	Develop java program to showcase the usage of this, static and final keywords	2#	CO2
II	7	Develop java program to demonstrate the concept of Wrapper classes, nested class, and inner class	2#	CO2
III	8	Develop java programs to illustrate the concept of different types of inheritance, use super keyword, method overriding, use final keyword in Java	2#	CO3
III	9	Develop java programs to demonstrate abstract class, abstract method, dynamic method dispatch	2#	CO3
III	10	Develop programs to demonstrate concept of interface implementation and extending interface, visibility control in inheritance hierarchy and across package	2#	CO3
IV	11	Develop java programs to demonstrate exception handling using try-catch-finally, throw, throws	2#	CO4
IV	12	Develop java programs to demonstrate exception handling using multiple catch, throw-throws constructs and new Exception handling features	2#	CO4
V	13	Write Java programs to demonstrate creation of thread using Runnable and Thread	2#	CO5

V	14	Write Java programs to demonstrate demonstrate multithreading, synchronization, isAlive(), join(), interthread communication, deadlock, suspending, resuming and stopping thread	2#	CO5
VI	15	Write Java programs to perform different file handling operations like reading, writing, copying files	2#	CO6
VI	16	Write Java programs to perform I/O operations using various stream classes, PrintWriter, Scanner, Object Serialization	2#	CO6

Note: # Compulsory

VIII Self-Study Learning (SLO in Cognitive/Psychomotor/Affective Domain)

1. Develop a Java program for dynamic data type conversion based on user input, facilitating conversion between different primitive data types
2. Develop a Java program for analyzing user inputted arrays, calculating statistical measures like mean, median, mode and standard deviation
3. Design an employee management system where different types of employees are handled uniformly
4. Develop an ATM transaction simulator with robust error handling
5. Implement a concurrent task executor for parallel execution of tasks
6. Develop a text file analyzer for analyzing word occurrences and character counts
7. Design a shopping cart system for an e-commerce platform, using classes and interfaces to manage items and calculate the total cost with discounts.
8. Create a file-based Java application to track personal expenses. Implement serialization to store and retrieve data.
9. Write a Java program to organize images from a folder into subfolders based on their extensions (e.g., .jpg, .png). Use file handling for implementation.
10. Implement a system to input student scores, calculate grades, and save the results in a file. Use abstract classes and exception handling to manage errors.
11. Develop a program to search for files by name or content within a specified directory. Incorporate error handling and file I/O operations.
12. Create a multithreaded application to simulate banking operations like deposits, withdrawals, and account balance checks.
13. Develop an application to manage library records, including book borrowing and returning. Implement object-oriented concepts like inheritance and interfaces.

IX Specification Table for Classroom Learning Assessment:

Unit No.	Units	Classroom Learning Hours	C/O	Levels from Cognition Process Dimension			Total Marks
				R	U	A	
1	Foundations of Java Programming	10	C	02	04	04	10
			O	02	00	04	6
2	Java Object Oriented Programming Essentials	12	C	02	04	10	16
			O	00	08	00	8
3	Inheritance, Polymorphism and Package Management	12	C	00	04	06	10
			O	00	00	06	6
4	Exception Handling	8	C	06	00	04	10
			O	02	04	00	6
5	Multithreaded Programming	8	C	02	04	06	12
			O	00	00	06	6
6	Java File Handling	10	C	02	04	06	12
			O	00	04	04	8
Total		60	C	14	20	36	70
			O	4	16	20	40

*C - Compusory O - Optional
*R – Remember U – Understand A – Analyze / Apply

X Question Paper Format for Summative Assessment (SA):

Q. No.	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Options	
	TLM	TLM	TLM	TLM	TLM	TLM	TLM	C	O
1	1R2	2R2	4R2	5R2	6R2	4R2	1R2	5	2
2	1U4	2U4	4R4	4U4	2U4			3	2
3	1A4	2A4	3U4	1A4	6A4			3	2
4	4A4	5U4	6U4	6U4	2U4			3	2
5	2A6	3A6	5A6					2	1
6	6A6	5A6	3A6					2	1

*T- Unit/Topic Number L- Level of Question M- Marks
*R – Remember U – Understand A – Analyze / Apply
*1R2 means Unit Number No- 1, Level of Question –Remember, Marks – 2 Marks

XI Scheme of Laboratory Formative Assessment (FA):

S.N.	Criteria	Max. Marks
1	Writing program logic	5
2	Program execution	10
3	Error debugging	5
4	Viva Voce	5
	TOTAL	25

XII Scheme of Self-Learning Summative Assessment (SA):

S.N.	Criteria	Max. Marks
1	Writing program logic	5
2	Program execution	10
3	Error debugging	5
4	Viva Voce	5
	TOTAL	25

XIII COs-POs/PSOs Mapping Matrix:

Course Outcomes	Program Outcomes							Program Specific	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO1	3	-	-	3	-	-	-	3	-
CO2	3	-	-	3	-	-	-	3	-
CO3	3	-	3	3	-	3	3	3	3
CO4	3	-	3	3	-	3	3	3	3
CO5	3	-	3	3	-	3	3	3	3
CO6	3	-	3	3	-	3	3	3	3

XIV Textbooks, BIS Codes References:

S.N.	Title	Author, Publisher, Edition and Year of Publication	ISBN Number
1	Java: The Complete Reference	Herbert Schildt, McGraw Hill,Eleventh edition, 2020	978-9390491629
2	Introduction to Java Programming	Y. Daniel Liang, PHI, 10th edition, 2013	978-0133761313
3	Head First Java	Kathy Sierra, Bert, Trisha Gee O'Reilly Media; 3rd edition, 2022	978-1491910771

XV e-References:

1. https://docs.oracle.com/javase/tutorial/ accessed on 4th August 2024
2. https://introcs.cs.princeton.edu/java/home/ accessed on 4th August 2024
3. https://docs.oracle.com/en/java/index.html accessed on 4th August 2024
4. https://math.hws.edu/javanotes/ accessed on 4th August 2024

XVI List of Major Equipment/Machineries with Specification:

1. Processor requirements: minimum a Pentium 2 266 MHz processor.
2. Memory Requirements: On Windows 64-bit operating systems, in 32- or 64-bit mode, the Java runtime requires a minimum of 128MB of memory.

XVII List of Industry Experts and Faculties who contributed for this curriculum:

S.N.	Name	Designation	Institute / Industry
1	Dr. M. V. Sarode	Head of Department, Computer Engineering	Government Polytechnic, Nagpur
2	Mr. S. A. Khatri	Lecturer, Computer Engineering	Government Polytechnic, Nagpur
3	Dr. U. B. Aher	Lecturer, Computer Engineering	Government Polytechnic, Nagpur
4	Dr. K. L. Bawankule	Lecturer, Computer Engineering	Government Polytechnic, Nagpur
5	Mr. Dipak Dhote	Operations Manager	IT-Network Infosystems Pvt LTD, Nagpur
6	Dr. Rakesh Kadu	Assistant Professor, Information Technology	Ramdeobaba College of Engg, and Management, Nagpur
7	Mrs. V .A. Raje	System Analyst	MSBTE,RO Nagpur

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